Designing a Research Setup to Measure Co-contraction with Affordable EMG

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Introduction

Since the early development of medicine, surgery has been considered one of the most complex disciplines. For many centuries, the training of surgeons was based on the principle of observation and practice on real patients. The student first observed the work of an experienced surgeon, and then gradually moved on to performing these actions under supervision. However, the problem with this approach was that any mistake could cost the patient their health or even their life. Mistakes were inevitable during the learning process. Since the practical experience of surgical students was based on real patients, the risk of making mistakes was high and heavily dependent on the careful observation of the surgeon's actions. As a result, this approach to training is no longer considered acceptable under modern safety standards (Shahrezaei et al., 2024). An operating error cannot be considered merely a technical aspect of training, as it directly affects the outcome of treatment and patient safety. At the same time, while theoretical knowledge and decision-making can be assessed through exams, evaluating motor skills remains a significantly more challenging task. To address these challenges and reduce risks to patients, alternative training methods have been developed, with simulation becoming one of the most prominent approaches.

Studies show that the use of simulators significantly improves surgeons' skills and reduces the number of errors, increasing the overall effectiveness of treatment (Fried et al., 2004). For example, it has been shown that trainees who use simulation perform more steps of the procedure and make 61% fewer critical errors compared to traditional training (Mascheroni et al., 2024). Thus, simulation allows not only for safe training, but also for isolating individual components of complex surgical tasks and observing the consequences without risk to the patient. This is especially true for modern VR simulators, such as LapSim, which allow for the reproduction of complex procedures while simultaneously recording accurate performance metrics. Unlike traditional training, where assessment is often limited to observation by an experienced surgeon, such simulation provides an opportunity to accurately measure execution time, error rates, and accuracy of actions, among other metrics. However, despite these metrics, they only reflect the outcome of the task, but do not show how the task is performed. For example, different participants may achieve similar results but use different strategies. One person may perform the movement more smoothly, while another may perform the movement with more tension and less control.

At early stages of learning, people tend to show more muscle coactivation, as they try to stabilise their movements and improve control (Darainy & Ostry, 2008). Therefore, it is

important to consider not only the outcome, but also the process of performing the movement. To capture this, physiological measurements such as electromyography (EMG) can be used, as they provide information about muscle activity during task execution. In this study, muscle activity is recorded using surface EMG electrodes placed over the forearm muscles, including the extensor digitorum and antagonistic flexors such as flexor digitorum superficialis and palmaris longus, following SENIAM guidelines (Hermens et al., 2000). Since the EMG signal often contains noise, such as electrical interference or electrode movement, the data is filtered (De Luca, 1997) to ensure that only the muscle signal is analysed (Clancy et al., 2002).

The study aims to provide an accessible experimental setup to measure forearm muscle co-contraction during laparoscopic simulation using surface EMG, and to present these results visually as real-time feedback. Specifically, it is investigated whether co-contraction defined as the percentage of trial time during which both the agonist and antagonist muscle groups simultaneously exceed a threshold derived from individual baseline activity is associated with LapSim performance, and whether it changes across five consecutive trials in a way that reflects motor learning. Two hypotheses were formulated. First, participants with higher forearm muscle co-contraction will achieve lower LapSim efficiency and tissue handling scores, reflecting greater neuromuscular stiffness and less refined motor control. Second, across five consecutive trials, the co-contraction index will decrease as LapSim performance scores improve, indicating progressive motor adaptation and increasing neuromuscular efficiency. This research aims to aid surgical coaches and tutors by providing a cost-effective, quantitative way to measure and visualise muscle activation during simulation training. An accessible substitute to the LapSim simulator using the same measurement setup has been provided in the methods section.

Methods

Experimental procedure

Participants: Co-contraction of the forearm was measured using a sEMG during a simulator task. Participants from varying levels of expertise (novice, expert) were instructed to perform a simulated laparoscopy surgical task using the LapSim simulator. This inclusion of different expertise levels was optional and serves primarily to explore variability in muscle activation rather than to evaluate performance differences. We aimed for three participants

who will each perform one experimental block of five trials. Participants were aged between 22 and 25 years old and were derived from the Human Factors and Engineering course at the University of Twente. None of the participants had experience in surgical task environments, although one participant was proficient in woodworking and therefore was able to perform hand-precision movements.

LapSim simulator task procedure: After the right program/task in the simulator environment has been selected (Basis Skills - Easy settings: Instrument Navigation), the participant was instructed to stand in front of the simulator and was given instructions on how to perform the simulated laparoscopy surgical task. Participants were instructed to grab the two surgical shear instruments in front. They were able to grab the instrument using the middle finger and thumb, almost the same way you would hold scissors in your hand. With their index finger, participants were able to rotate the jaw/tip of the instrument via a gear on top of the scissor-like instrument. To start the experiment, the instruments first had to be calibrated on movement and position. This was done by setting the instruments upright in their holders, after which the participants could raise the instruments vertically to automatically start the trial. Before the start of the experiment, a baseline phase of ten seconds was measured where participants held their arms still whilst holding the instruments in their resting state. After lifting up the instruments and starting the experiment, the participant would see a virtual environment on screen representing tissue inside of the body, including the two instruments the participant is holding. The participant should aim to keep these instruments inside the window frame of the screen. For the task, the participant was instructed to cautiously poke tissue damages - yellow tumor-like dots appear on screen, one at a time at random positions, for 10 times. One of the instruments (left or right) would unobtrusively light up in green to indicate which instrument should be used to remove the tissue damage.

During the task, efficiency and tissue handling were automatically measured within the LapSim program. For efficiency, it measured the time to find all targets, the path length from one target to the next, and the angular path degrees, both for the left and right instrument. For tissue handling, the amount and size of tissue damage, and the amount of occurrences and total time the instrument was outside of view, were measured for both the left and right instruments. Efficiency and tissue handling produce separate performance sheets on which a feedback bar was presented for each variable. The feedback bar consisted of three regions depicted in red, orange and green, with red indicating poorer performance and green indicating good performance. This determination of performance rate was based on

a benchmark method, in which performance was normalized relative to previously recorded expert-level performance, with expert performance representing the upper bound (100%) on the feedback scale. However, this meant that there was no way to detect progression if performance was at 100%, when we solely relied on the LapSim performance results.

Measurement setup

Apparatus: For our setup, we used the Ylab pro software environment provided by Martin Schmettow in combination with the STM32 Nucleo F446ZE board (Appendix 2/Documentation). The board has two micro usb ports, one for power supply and one for data transfer and is able to process data at a maximum of 250 Hz. Furthermore, Ylab pro is a mandatory program which should be downloaded to run the Y-studio zero program, which allows for (s)EMG measurement. The Y-studio zero was taken from the Github repository by Martin Schmettow, but needed some altering in order to function for the current setup.

sEMG: Muscle activity was recorded using surface electromyography (sEMG) in order to measure co-contraction as a proxy for stiffness of the forearm. Bipolar surface electrodes were placed on the extensor digitorum muscle group (agonist, X, dorsal), and the opposing flexor digitorum superficialis and palmaris longus (antagonist, Y, ventral). These particular muscles have been chosen because; (1) these are big muscles, ensuring a strong signal; (2) they lay on the surface, ideal for sEMG; (3) and are separated clearly, which is beneficial for calculating co-contraction. The electrodes should be placed in a mid-belly position, avoiding tendon regions. That is, the electrodes should be placed on the thickest part of the muscle to reduce noise artifacts since the EMG signal is the strongest at this location. The orientation of the electrodes should be parallel to the muscle fibers, thereby maximising signal consistency since EMG signals spread through the muscle fibers. The electrodes were placed at approximately 20mm centre-to-centre (metal measuring points) distance. Electrode set 1, containing sensors marked 1 - 3 were placed on the agonist muscle on the top and bottom of the muscle, with reference electrode 3 placed on the bone area of the elbow. Electrode set 2, containing sensors marked 4 - 6 were placed on the antagonist muscle similarly, with the reference electrode 6 again placed on the bone area of the elbow. The electrode size was 10mm. Electrode placement has been based on presentation slides of Russel Chan and established guidelines such as SENIAM. Prior to electrode placement, the skin was treated and gel was added to reduce input impedance and improve signal quality.

The “differential” (or bipolar, or Single Differential – SD) detection is useful because unwanted common mode input signals such as the power line interference (PLI), are theoretically cancelled at the detection point.

Data acquisition

Data processing sEMG: Before the start of the measurement, it was required to iteratively select the particular sensor and its corresponding signal number in the Y-studio Zero program. The electrical activity generated by muscle contractions was detected by the surface electrodes as an analog signal. This signal was amplified and conditioned by the EMG system. The conditioned signal was then sampled at a fixed sampling rate at 250 Hz (due to hardware constraints) and converted into a digital signal through analog-to-digital (A/D) conversion. The resulting time-series data represented the raw sEMG signal for each recorded muscle during task execution.

The raw EMG signals were then processed using standard signal processing to reduce noise and filter our relevant muscle activity. The EMG program (Y-studio zero) allowed us to adjust the low-pass filter accordingly within the program. First, a band pass filter of 20 - 450 Hz is applied. The filtered signal was then full-wave rectified to obtain the absolute value of the signal. Subsequently, a low-pass filter of 5 Hz was applied (in the Y-studio Zero program) to produce a smoothed representation of the signal.

After processing, the data was acquired in a ylb file and consists of 5 columns. The data per participant per trial (including baseline) was produced in separate files. The first column represents the timestamp of when the sEMG measurement starts. Second (and also the third column for unknown technical reasons) is the apparatus identity marker - indicating the sEMG or the IMU input. Fourth is the marker for which electrode the sEMG value results from. The fifth column is the sEMG value itself. The code separates the sEMG values of electrode 1 - 3 (muscle 1 - agonist) from the sEMG values of electrode 4 - 6 (muscle 2 - antagonist), in order for us to easily measure co-contraction. Ultimately, one file per participant per trial was produced; participant - baseline, participant - trial 1, etc.

Data analysis, co-contraction, feedback visualisation and performance measure: These files were then processed in another Python based code, see Appendix 2/Code for the full code and Appendix 3 for some excerpts of AI assistance in the creation of this code. It searches through the appointed folder for the file containing the baseline value and the file containing a certain trial value for a specific participant (participant x baseline & participant x

trial) per set of electrodes. The name of this folder within the code should be adjusted accordingly, as mentioned in notes within the code. For the baseline measurement, the mean and standard deviation were saved for each participant per set of electrodes. Afterwards, the trial values of the participant in question were compared to the baseline values to see when the sEMG values of both sets of electrodes exceeds the mean baseline value + 2 * the standard deviation of the baseline value. Remember that co-contraction was defined as the simultaneous activation of opposing muscles as a proxy for arm-stiffness. Therefore it is important that the sEMG values of both sets of electrodes meet these requirements. The percentage of time per trial the sEMG value met these conditions was printed. With this procedure, a percentage was obtained for the time per trial that there was higher co-contraction present compared to the baseline in the forearm.

This procedure was then repeated for each of the five trials (still within the same test block using the same initial baseline measurement). This percentage was then used as a feedback visualisation measure. The percentage of time per trial for when the sEMG value met the criteria was compared for each of the five trials against the previous one. A visual prototype for a website was also made to create a clearer view of the feedback for the participant (Appendix 3) with the function of a visual feedback diagram. This visual prototype was heavily inspired from fellow students after receiving verbal permission to do so. The value was presented visually on screen, and allowed for easy comparison between trials and see for progress between the trials of each block.

Additionally, the LapSim simulator task results, consisting of efficiency and tissue handling, were compared to previous trials. Efficiency and tissue handling within the LapSim program were determined using the available benchmark performance feedback visualisation (see “Simulator task procedure”). Iteratively, researchers looked at the amount of performance rates in the red v.s. orange v.s. green area, and loosely determined performance rating on the simulator task. After taking into account both the sEMG and the IMU results, including the simulator performance results, the participant was informed about his performance and was given recommendations for the following trial. For instance, the participant was recommended to “try to relax a bit for the next trial” when higher stiffness (co-contraction) and jerkiness (acceleration) in combination with lower simulator performance are detected.

Synchronisation of measurements and task: Since the provided code was difficult to adjust because of technical limitations and it was impossible to invade the LapSim environment, the sEMG measurement for baseline and task were rudimentarily synchronised

to the start and end of each trial by manually pressing start and stop on the measurement tools. Measurements were started within one second prior to task onset. Precisely, the measurement of the sEMG was started simultaneously one second before the start of the LapSim simulator task to allow for margin to let the program start up. This is done by one of the experimenters on-site.

Human Factors and replication

Human Factor Design consideration: It should be mentioned that the sEMG setup comes with the problem of wires that bungle from the point where the wires are attached to the sensor on the skin of the participant to the point where they are connected to the STM32 board. Loosely bungling wires could impede participant's performance by limiting range or reducing concentration. Therefore, to account for usability of this setup, the researchers came up with the following solutions. First of all, simple arm sleeves have been added to the setup. Multiple common runner's phone bracelets or a runner's armguards could be used to hold the wires in place from the point of sensor contact in order for them to connect at an end-point of the sleeve and then proceed the route down to the board in an unobtrusive way that does not impact movements of the participant. However, this setup still allows for the wires to drop downward from the end-point where they connect at the end of the sleeve to the point where they connect to the STM32 board. That is, the wire setup at the arm is organised, but they are still bungling from that end-point. Therefore, a stance or high table could be placed next to the participant at elbow height whilst performing the task. The STM32 board and the laptop could be placed on this higher plateau. This lesser height difference disenables the downward falling of the wires. Additionally, simple cable clip organisers or clothespins could be used to direct the wires to this higher plateau and ensure as little movement - and therefore noise artifacts - of the wires as possible.

This setup allows for a cost-effective, easy to implement and organised way to measure quantitative muscle activation data and to present it visually using sEMG. See Appendix 2/Documentation for an actual illustration of the described setup.

Recommendation for replication: For replicability of this setup, it should be taken into account that seldom people have access to the LapSim simulator. To compromise the inaccessible LapSim simulator, the surgical task is easily replicated using a simple makeshift box simulator based on the Inovus Medical box simulator. This setup consists of a big plastic box containing several items to be manipulated by the instruments. The surgical instruments are inserted into two holes that have been cut or drilled into the lid of the box. This cut-area

has been reinforced using glue in order to lessen the chance of breaking or tearing the lid during manipulations. The items that are to be manipulated consist of simple household sponges that have been cut into small pieces - 6 pieces in our case. The items are laid out on the left side of the box simulator, laid out in a 2x3 grid. On the right side, there are 6 sections that have been marked out on the surface. The task of the participant was to bring the items piece by piece over to the right side using the left instrument to pass the item over to the right instrument and place it in a particular marked section.

With this setup, unlike the LapSim simulator, it is possible to increase the difficulty of a particular task. Using the LapSim simulator it is possible to select different difficulty levels, although this introduces a new task that is different from the Instrument Navigation task. Initially, the participant was instructed to place the items on the left in a marked out section on the right free of choice. To increase difficulty, the participant received instructions on which particular section to place the item, one by one. To include a second task to induce cognitive load, the indications on which particular section to place the item could be given before the start of the task instead of with every item. Although the surgical task of the box simulator does not quite represent the virtual environment of the LapSim simulator task, this setup allows for cost-efficiency, accessibility, and flexibility by providing an innovative way to implement different levels of difficulty to the surgical task.

Validation of results

To check the reliability of our EMG sensor in accurately measuring co-contraction, some analyses needed to be done. Once the data files of the participants were correctly named, the code to measure co-contraction was run. This analyses, comparing the values from the trial by seeing if the values are above the baseline mean + 0.1 * the baseline standard deviation, indicated co-contraction within each trial (Table 1). We have deferred from the original formula (baseline_M + 2.0 * baseline_SD) since we were unable to find substantial differences in co-contraction during the task compared to baseline measurement. The reason for adjusting the formula has been explained at the end of the results section.

Table 1

Co-Contraction % per participant per trial

Participant	Trial	% of time Co-Contracted
1	1	5.79
1	2	7.70
1	3	7.15
1	4	8.23
1	5	4.72
2	1	1.35
2	2	1.35
2	3	1.22
2	4	2.29
2	5	1.58
3	1	8.01
3	2	2.86
3	3	4.09
3	4	6.98
3	5	9.91

Basic Analyses

Participant	Min %	Max %	Mean %	SD %
1	4.71	8.22	6.72	1.441
2	1.22	2.29	1.56	0.429
3	2.86	9.91	6.37	2.876

Table 2 presents co-contraction values for all three participants across five trials. Participant 2 showed the lowest and most stable co-contraction (mean = 1.56%, SD = 0.43), consistent with their high LapSim performance. Participant 1 showed moderate values (mean = 6.72%, SD = 1.44) with no clear trend across trials. Participant 3 had the highest variability (mean = 6.37%, SD = 2.88), again without a monotonic decrease over trials. Overall, the expected learning-related reduction in co-contraction did not emerge within five trials for any participant. To examine this relationship more formally, linear models were estimated in R.

Table 2

Muscle mean + SD per participant per trial

participant	trial	muscle1_mean	muscle1_std	muscle2_mean	muscle2_std
1	1	1598,60	255,305	1655,59	202,627
1	2	1599,31	254,234	1656,80	202,095
1	3	1599,64	253,850	1656,91	201,918
1	4	1599,22	254,426	1656,69	201,858
1	5	1599,39	254,248	1656,36	201,439
2	1	1598,92	254,145	1656,39	201,502
2	2	1599,28	254,851	1656,04	201,481
2	3	1599,44	254,217	1656,550	201,381
2	4	1599,09	254,536	1656,19	201,428
2	5	1599,99	253,377	1657,24	200,730
3	1	1600,63	254,986	1657,60	201,945
3	2	1601,42	255,437	1657,02	201,131
3	3	1601,67	256,072	1657,15	201,587

3	4	1601,56	256,902	1657,17	201,721
3	5	1607,77	265,310	1656,59	202,844

After completing the first trial, participant 1 mentioned that they were very focused on the LapSim task itself and therefore were not fully aware of the presence of co-contraction or "stiffness" during the task. After the following 4 trials however, participant 1 did mention that they felt like they experienced some co-contraction because "they were trying very hard to do well". In contrast, participant 2 stated after each trial that they were pretty relaxed while performing the simulator task and that they themselves did not really notice any co-contraction during the trials. Participant 3 said that they were relatively anxious while performing the first trial, as they were totally inexperienced with the simulated task. After their first trial, they mentioned being more relaxed but wanting to do better on subsequent trials as they were disappointed with their scores, and also did mention feeling more stiff because of this.

The self-reported co-contraction by the participants, or as they called it "stiffness", is in line with the results from the EMG analyses. Participant 1 had more mentions of co-contraction after their trials compared to Participant 2 which is also what the EMG results indicated. Results indicated the highest overall co-contraction within participant 1 with a mean of 6.72, while participant 2 experienced the lowest co-contraction across trials with a mean of 1.56. Participant 3 mentioned varying levels of co-contraction, which the EMG results also suggests as their co-contraction levels start off high in trial 1, then decrease in trial 2 and start to increase again in the following trials.

However, looking at the raw values of the EMG raised doubts about its accuracy (Table 2). The means of the muscle electrodes showed very high values across all trials, but most of all very high standard deviation all being around 250 for muscle 1 and 200 for muscle 2. Because of these doubts surrounding the raw values, the baseline values from Participant 1 were compared to itself to check if the calculation method can be trusted. This calculation suggested a co-contraction of 1.12% within the baseline itself.

Discussion

This study investigated whether forearm muscle co-contraction, measured using surface EMG, is associated with laparoscopic simulation performance and whether it decreases across five trials as participants gain experience. Based on the statistical results, none of the hypotheses was supported and it is important to consider why the results turned out this way.

The first issue is the small sample size. With only three participants and fifteen observations in total, the statistical analyses had very limited power to detect a real effect even if one existed. Related to this, the between-participant differences are large. Participant 2 scored between 90% and 100% across all five trials from the very beginning. This means there was almost no variation in performance for this participant, which makes it impossible to detect a correlation with co-contraction. For Participant 2 specifically, a ceiling effect applies. Participant 1 showed more variation in scores, ranging from 73% to 100% and failing two trials, but their co-contraction also did not change in a consistent direction. Participant 3, who improved most clearly from 26% to 100%, showed the most interesting pattern. However, their co-contraction did not decrease across trials as the first hypothesis would predict. In fact, the highest co-contraction value for Participant 3 was recorded in Trial 5, the trial with the highest LapSim score. This is the opposite of what was expected, and it is difficult to attribute this simply to measurement error. The trending result for pass/fail status ($p = 0.086$) offers some weak support for the first hypothesis, but three participants and fifteen observations are not enough to draw confident conclusions.

The second hypothesis predicted that co-contraction would decrease as LapSim performance improved across trials, reflecting motor learning. This was not supported either. Participant 2 showed consistently low co-contraction throughout all five trials, which is consistent with their high performance level, but since both co-contraction and performance remained stable, there is no learning trend to observe. Participant 3, who showed the clearest improvement in performance, did not show a corresponding decrease in co-contraction. Instead, the values went from 8.01% to 2.86% to 4.09% to 6.98% and finally 9.91%, with no clear direction. It is possible that five trials within a single session is simply not enough time for the neuromuscular system to adapt in a way that is visible in the co-contraction index. Motor learning at this level may require more repetitions or multiple sessions before consistent changes can be measured.

Most importantly, the experimenters were unable to adhere to the agreed formula for calculating co-contraction (baseline_M + 2.0 * baseline_SD), and adjusted the multiplication to 0.1. This however, despite being able to account for the differences in activation during the

task compared to baseline, is not a reliable or legitimate method for calculating co-contraction. Reduction in reliability may be due to measuring false-positives for muscle activations, or an erroneous baseline measurement due to incomplete relaxation.

There were also practical limitations that could have affected the results. The sEMG recording had to be started manually at the beginning of each trial, which introduced a timing difference of approximately one second. Because the co-contraction index is calculated as a percentage of total trial time, this timing margin could cause small errors in the result. Additionally, all three participants were students without any surgical experience. This makes it difficult to know whether the findings would apply to an actual clinical training context. It was also not possible to compare the participants to an experienced reference group, which would have helped to establish what a low or high co-contraction level means in practice. A comprehensive and chronological description of the challenges and limitations including their subsequent solutions is given in Appendix 1.

Despite these limitations, the study shows that measuring co-contraction using surface EMG during simulated laparoscopy is technically feasible with relatively simple and low-cost equipment. The between-participant pattern is worth noting: Participant 2, who performed best, also showed the lowest and most stable co-contraction, while Participant 3, who struggled most in the early trials, showed the highest co-contraction values overall. This difference is consistent with the theoretical expectation that skilled performers use their muscles more efficiently. Without significant results however, the authors are unable to advocate against the theoretical framework. The most useful contribution of this project is the real-time feedback interface, which shows participants their co-contraction percentage after each trial. It gives coaches and participants specific physiological information that the LapSim score alone does not provide. Moreover, the methodology describing the box-simulator setup substituting the LapSim simulator is beneficial for cost-effective replication

Future studies should include more participants with different levels of surgical experience, so that a wider range of performance and co-contraction values can be observed. A multi-session design, spread over several weeks, would give motor learning more time to produce measurable changes. Automated synchronisation between the sEMG recording and the simulator task would reduce timing uncertainty. Finally, examining co-contraction in more specific muscle groups, rather than treating the agonist and antagonist sets as single units, might reveal more detailed patterns of how forearm muscle control changes with practice.

Conclusion

This study demonstrated the technical and practical feasibility of measuring forearm muscle co-contraction using surface EMG during a simulated laparoscopy task on the LapSim simulator. A co-contraction index was successfully computed for three participants across five consecutive trials, and the results were presented in a real-time visual feedback interface designed to support coaching and self-monitoring during simulation training.

The hypothesised associations between co-contraction and LapSim performance were not confirmed statistically within this small-sample study. A ceiling effect in LapSim scores, a limited number of trials, and a small participant group were the primary factors constraining the ability to detect such associations. Nevertheless, the between-participant pattern with the most experienced participant showing markedly lower co-contraction than the others provides directional support for the theoretical premise that neuromuscular efficiency distinguishes skilled from unskilled performance in surgical simulation.

The experimental setup developed here combining the STM32 Nucleo board, Y-studio zero software, surface electrodes on the forearm, and a web-based feedback visualisation is low-cost, replicable, and non-invasive. It constitutes a useful starting point for further research into physiological measurement during laparoscopic simulation training, and the real-time co-contraction feedback component represents a concrete, practically actionable contribution to surgical coaching methodology.

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Appendix 1

Challenge-solution logbook

Trying out a novice setup using sEMG and IMU inevitably introduced technical and practical challenges. Here, we present our logbook on which challenges and problems we encountered, how this has been approached and what solution has been found.

Week 1: During the first week of our project, we directed most of our attention to aligning the project with our stakeholder. The assignment was to create a way to quantitatively measure muscle activity and be able to present the performance results visually as a feedback scale. In combination with another stakeholder, it was suggested to apply this setup to a laparoscopy task. Iteratively, it was proposed to measure stiffness of the forearm during a laparoscopy simulator task using sEMG, and to visually present it as a feedback scale for easy and quick analysis during task execution.

On top of that, a theoretical challenge was introduced. The intention was to measure stiffness of the forearm, as poor or novice performers were thought to have higher amounts of stiffness during task execution. Therefore, it was required to argue the proxy of stiffness, as to how this is measured quantitatively. Eventually, the authors were able to argue the proxy of stiffness using the co-contraction index.

The main achievement of the first week was to have a clear and coherent proposal of the project, in which we measured stiffness of the arm using sEMG to measure performance quantitatively during a surgical simulator task.

Week 2: During the second week, the authors had their first meeting with the primary stakeholder, where they pitched the idea. After discussing the idea with the stakeholder, several technicalities were introduced that had to be reviewed. Particularly, the authors and the stakeholder discussed the concept of noise artifacts, and how easily these are introduced into the signal. Using this knowledge, the authors intensively reviewed literature on how to reduce noise artifacts during (s)EMG measurements and which noise artifacts should be taken into account. After review, the authors were able to create an sEMG envelope that was suitable for our measurement.

Furthermore, the stakeholder introduced the human factors perspective behind our project. After attending the authors to the problem behind this setup - the loosely bungling wires - the authors figured out an organised setup using an arm sleeve, high table and cable clips - as discussed in the methods section. Fortunately, the experimenters felt comfortable

with this requirement, and were able to come up with an approachable, innovative and cost-effective solution. Moreover, this perspective benefitted the experimenters with consolidating the scope of this project.

Week 3: During the third week, the authors were to present the pitch officially to the class and stakeholders. However, due to accessibility and availability, the authors still had not received any idea of information about the simulator task that was to be used. Therefore, the authors were able to present the main idea and goal of the experiment, the consolidated sEMG processing procedure, and the idea on the visual feedback presentation, but had to make up the task manipulation, which impeded the clearance of the aim of our project.

However, one of the stakeholder was dissatisfied with solely the implementation of an sEMG measurement, and advocated the use of an IMU MotionSensor. With consideration, the authors managed to theoretically figure out the implementation of the IMU MotionSensor, as the measurement of acceleration was used as a proxy for jerky movements to explain poorer performance. The IMU MotionSensor setup was later implemented physically.

Lastly, the authors received the STM32 board and the sEMG electrodes from one of the stakeholders, which allowed us to review the implementation of this setup, and allocate our attention to the technical part of our project, enabling vast progress for this project.

The main achievement of this week was, fortunately, to be able to figure out the data processing and theoretical grounds for implementing the IMU MotionSensor easily and quickly, benefitting further advancements on the progress of this project.

Week 4: During week four, the authors allocated most attention to finalising the code and figuring out the technical requirements behind this setup. The STM32 board and the Y-studio zero program introduced lots of crucial issues that had to be resolved. For instance, the code from the Y-studio zero program was written in Rust coding language and built for a Linux environment, whilst we were only partially proficient in Python and used Windows. After tedious research, the authors figured out a way to get this setup to work for a Windows environment. However, although we were able to measure sEMG correctly using this program, the authors were unable to adjust the code accordingly because of the coding language limitations. The authors bypassed this issue by implementing an additional Python based code to process the data accordingly (as described in the methods section), and were able to accomplish a correct way of measuring co-contraction of the forearm using sEMG.

A big advancement on the progress of this project was when the experiments finally were able to access the LapSim simulator for inspection and orientation. After becoming familiar with the simulator, the experimental task and the performance measures, the

experimenters were able to extend on and consolidate the experimental procedure. Moreover, after inspecting and understanding how the performance measures were constructed, the experimenters had a clear view of how the overall results of the setup should be interpreted, resulting in the combination of sEMG, IMU and the task performance results. Although the task performance results of the simulator task were limited by the benchmark method, these results were useful for easily detecting trial-by-trial performance progress.

Week 5: During week five, the authors were introduced with a practical availability problem. The project setup, including the human factors perspective, was essentially ready to be tested. However, due to inaccessibility of the LapSim simulator, the authors had to wait a long time before they were able to test the setup, receive and analyse their data. The authors compensated for this by preparing the methods procedure beforehand extensively, and finalised most of the project report except for the results and discussion section.

Moreover, in accordance with the stakeholder, the authors chose to use the three experimenters themselves as participants, as there already was seemable variety in hand-precision proficiency between the three of them, excluding the need to find real participants.

Week 6: Approaching the last week of this project, the authors had conducted the experiment to test the proposed setup on the LapSim simulator. Fortunately, the experimenters were able to test the setup positively, achieving interesting results. However, this was not without challenges. At first, due to the micro usb port being loosely mounted to the board, the Ylab program was initially unable to recognise the usb port. This could be resolved however by careful handling of the usb. Second of all, seemingly, the LapSim simulator did not have any usb port to export the task performance data directly to the laptop. Luckily, a staff member was able to create an account with one of the author's names, and was able to export the results in an excel sheet, which has been analysed and plotted by the experimenters. Third of all, the electrodes initially produced unexpected signals. However, after iteratively adjusting the electrode placement, the experimenters were able to retrieve a correct signal. Fourth, due to no prior experience with the Ylab-zero, the structure was unfamiliar. There was an issue with selecting the correct signals corresponding to the electrodes in the Ylab program. Using trial and error, it was possible to turn off the signal graph in the Ylab program - although this did not stop data acquisition of that electrode - and iteratively figure out the corresponding electrode by touching and pressing it to see the changes in the corresponding graph in the YLab. Last of all, most importantly, the EMG signal during muscle activation was far from the calculated threshold value mentioned in the

methods section. Since the EMG muscle activation signal rose 100-200 microvolts above the baseline value, the experimenters have chosen to adjust the calculation of the threshold to be the mean baseline value + 0.1 * the standard deviation of the baseline value, instead of 2.0, since a value of 0.1 was able to cover this change in muscle activation accurately. Remarkably, the EMG values sometimes declined below the baseline value, most likely because of noise artifacts created by the lower-quality electrodes or incorrect handling of the electrodes by the experimenters.

Additionally, after the experiment had been completed, the authors sat down to set up the visual feedback software. The feedback software was created for Linux and Apple applications. Although there were minor problems, the authors were able to quickly resolve these challenges and set up the feedback software correctly. One final challenge troubled the experimenters. Namely, implementing the IMU MotionSensor to the setup. However, as discussed in week 4, the authors endured the hardest challenge setting up the STM32 board, which caused unforeseen delay and hold of progress during our project. Fortunately, the experimenters were able to obtain the IMU MotionSensor, created a code to acquire the IMU data, and theoretically supported and explained the methods behind the implementation of the IMU MotionSensor for this particular setup.

In any case, we were unable to correctly test this setup, using sEMG to quantitatively measure stiffness of the arm during a surgical simulator task and, but were able to present these results visually in order to aid surgical coaches during training. Finally, the authors named their innovative setup: TenSim, appropriately derived from the words “tension” and “simulation”.

Appendix 2

Github repository

The following is a link to the main github repository from the project:

GitHub TSS

The following provides a link to the github repository containing the Documentation of the project:

GitHub Documentation

The following provides a link to the github repository with the code from the project:

GitHub Code

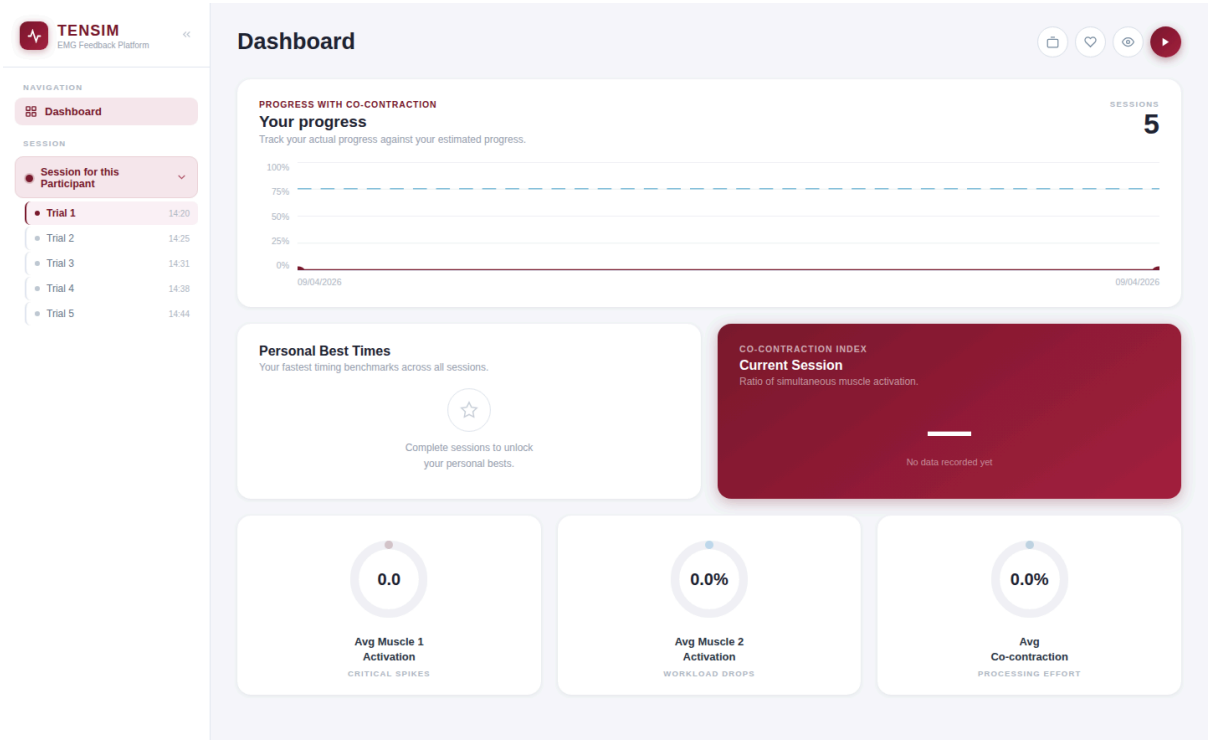
Appendix 3

Visual Prototype

Figure 1 provides a picture of a visual prototype for a website used for providing feedback to the participant after performing all the trials. This prototype contains the potential name of the setup: “TenSim”, a section on the left providing access to the co-contraction data from the different trials, a main graph indicating progress across trials, and in the bottom some further information on personal statistics like average muscle activation for muscle 1 and 2 and the average co-contraction.

Figure 1

Visual Feedback Prototype



AI statement

During the creation of this project, some AI has been used in the form of ChatGPT5.4. The writers of this report take full responsibility for the contents of the written work. The only used purpose of AI was that of providing assistance in coding for the y-studio zero and the EMG analysis code. Following are some excerpts from the conversation with ChatGPT that helped to give assistance when in need. AI helped to provide a starting point if the researchers did not know how to continue. The screenshot below provides an example from when the researchers were trying to create a filepath for the EMG analysis code to read the participant data files.

```
def load_emg_file(filepath):
    """Load EMG file and keep only relevant columns."""

    # Try automatic delimiter detection
    df = pd.read_csv(filepath, sep=None, engine="python")
```

Another example is when the AI was asked to provide an example of a working outcome of the EMG analysis code to check whether the example code given by ChatGPT was correctly helping the researchers to get the outcome they wanted, see below.

Example: 1 participant, multiple trials

Files:

```
Participant 1 Baseline.yld
Participant 1 Trial 1.yld
Participant 1 Trial 2.yld
Participant 1 Trial 3.yld
```

Output:

	participant	trial	muscle1_mean	...	muscle1_percentage_exceeding	muscle2_percentage_
0	1	1	...	...	8.5	12.3
1	1	2	...	...	10.1	15.0
2	1	3	...	...	6.7	9.8

At one point the researchers realised that the example code offered by ChatGPT only calculated muscle activation percentage for 1 muscle, and not for co-contraction. The screenshot below shows the resolution that the AI offered to fix this issue.

```
# Check condition: BOTH exceed threshold
condition = (combined["muscle1"] > m1_threshold) & \
            (combined["muscle2"] > m2_threshold)

percentage = (condition.sum() / len(combined)) * 100

return {
    "both_muscles_percentage_exceeding": percentage
}
```